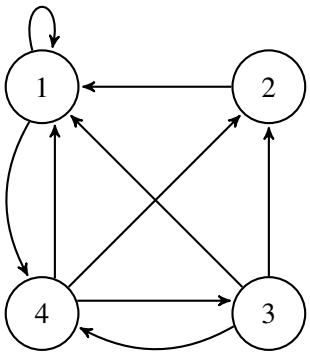
EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 7A

1. PageRank: Computing the Score Vector

Consider the network of four webpages below. An arrow from node j to node i means that page j links to page i.



Each page splits its visitors equally among the pages it links to, so the connectivity matrix is

$$\mathbf{C} = \begin{bmatrix} \frac{1}{2} & 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{3} & 0 \end{bmatrix}.$$

Let $\mathbf{s}[n] = \begin{bmatrix} s_1[n] & s_2[n] & s_3[n] & s_4[n] \end{bmatrix}^\top$ hold the fraction of all visitors on each page at time step n. At every step each visitor clicks a link, so

$$\mathbf{s}[n+1] = \mathbf{C}\mathbf{s}[n].$$

(a) Suppose the visitors start out spread evenly over the four pages: $\mathbf{s}[0] = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}^\top$. Compute $\mathbf{s}[1]$. What do its entries sum to, and why?

Answer: Multiplying $\mathbf{C}$ by a vector whose entries are all $\frac{1}{4}$ gives $\frac{1}{4}$ times the sum of each row of $\mathbf{C}$:

$$\mathbf{s}[1] = \mathbf{C}\mathbf{s}[0] = \frac{1}{4}\begin{bmatrix} \frac{1}{2}+1+\frac{1}{3}+\frac{1}{3} \\ \frac{1}{3}+\frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{2}+\frac{1}{3} \end{bmatrix} = \frac{1}{4}\begin{bmatrix} \frac{13}{6} \\ \frac{2}{3} \\ \frac{1}{3} \\ \frac{5}{6} \end{bmatrix} = \begin{bmatrix} \frac{13}{24} \\ \frac{4}{24} \\ \frac{2}{24} \\ \frac{5}{24} \end{bmatrix}.$$

The entries sum to $\frac{13+4+2+5}{24} = 1$. Each column of $\mathbf{C}$ sums to 1, so every visitor leaving a page lands on exactly one page: nobody enters or leaves the network. After just one step, page 1 already holds more than half of the visitors.

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(b) The PageRank *score vector* $\mathbf{s}^*$ is the distribution of visitors that no longer changes from one step to the next; this is the *equilibrium state* of the network. Write down the equation that $\mathbf{s}^*$ must satisfy. What does this say about $\mathbf{s}^*$ in terms of the eigenvalues and eigenvectors of $\mathbf{C}$? In terms of a null space?

Answer: If the distribution does not change after one more step, then

$$\mathbf{C}\mathbf{s}^* = \mathbf{s}^*.$$

This is the eigenvector equation $\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$ with $\lambda = 1$, so $\mathbf{s}^*$ is an eigenvector of $\mathbf{C}$ associated with the eigenvalue $\lambda = 1$. Moving everything to one side,

$$(\mathbf{C} - \mathbf{I})\,\mathbf{s}^* = \mathbf{0},$$

so $\mathbf{s}^*$ lies in the null space of $\mathbf{C} - \mathbf{I}$. Since the entries of $\mathbf{s}^*$ are fractions of the visitors, we also need them to sum to 1; this picks out one vector from the null space.

(c) Find the score vector $\mathbf{s}^*$ for this network.

Answer: We solve $(\mathbf{C} - \mathbf{I})\mathbf{s} = \mathbf{0}$:

$$\begin{bmatrix} -\frac{1}{2} & 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & -1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & -1 & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{3} & -1 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \end{bmatrix} = \mathbf{0}.$$

Rows 2 and 3 are already in a convenient form, so we can work upward with s_4 as the free variable:

$$\begin{aligned} &\text{Row 3:} \quad s_3 = \tfrac{1}{3}s_4 \\ &\text{Row 2:} \quad s_2 = \tfrac{1}{3}s_3 + \tfrac{1}{3}s_4 = \tfrac{1}{9}s_4 + \tfrac{1}{3}s_4 = \tfrac{4}{9}s_4 \\ &\text{Row 4:} \quad \tfrac{1}{2}s_1 = s_4 - \tfrac{1}{3}s_3 = s_4 - \tfrac{1}{9}s_4 = \tfrac{8}{9}s_4 \quad \Longrightarrow \quad s_1 = \tfrac{16}{9}s_4 \end{aligned}$$

Row 1 gives no new information: $-\frac{1}{2}\cdot\frac{16}{9} + \frac{4}{9} + \frac{1}{9} + \frac{1}{3} = \frac{-8+4+1+3}{9} = 0$. Choosing $s_4 = 9$ to clear the fractions,

$$\mathbf{s} = \alpha \begin{bmatrix} 16 \\ 4 \\ 3 \\ 9 \end{bmatrix}, \quad \alpha \in \mathbb{R}.$$

The entries sum to 32α, so we need $\alpha = \frac{1}{32}$:

$$\mathbf{s}^* = \frac{1}{32}\begin{bmatrix} 16 \\ 4 \\ 3 \\ 9 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{8} \\ \frac{3}{32} \\ \frac{9}{32} \end{bmatrix}.$$

(d) Rank the four pages from most to least important. Pages 2 and 4 each have exactly two incoming links. Do they receive the same score? Explain what you see.

Answer: From most to least important: page 1 $\left(\frac{16}{32}\right)$, page 4 $\left(\frac{9}{32}\right)$, page 2 $\left(\frac{4}{32}\right)$, page 3 $\left(\frac{3}{32}\right)$.

Pages 2 and 4 do not receive the same score, even though both have two incoming links. Page 2 is linked from pages 3 and 4, each of which passes along only a third of its visitors. Page 4 is linked from page 3 and from page 1, and page 1 is the most visited page and sends half of its visitors to page 4:

$$s_4^* = \tfrac{1}{2}s_1^* + \tfrac{1}{3}s_3^* = \tfrac{8}{32} + \tfrac{1}{32} = \tfrac{9}{32}, \qquad s_2^* = \tfrac{1}{3}s_3^* + \tfrac{1}{3}s_4^* = \tfrac{1}{32} + \tfrac{3}{32} = \tfrac{4}{32}.$$

PageRank does not just count incoming links. A link from an important page, and from a page that has few other outgoing links, is worth more.

2. Diagonalization and PageRank

The PageRank *score vector* $\mathbf{s}^*$ of a network of webpages is the distribution of visitors that no longer changes from one step to the next: $\mathbf{C}\mathbf{s}^* = \mathbf{s}^*$, where $\mathbf{C}$ is the connectivity matrix and the entries of $\mathbf{s}^*$ sum to 1. In other words, $\mathbf{s}^*$ is an eigenvector of $\mathbf{C}$ associated with the eigenvalue $\lambda = 1$. In this problem we use *all* of the eigenvalues and eigenvectors of $\mathbf{C}$ to understand why visitors settle into the score vector, and how quickly.

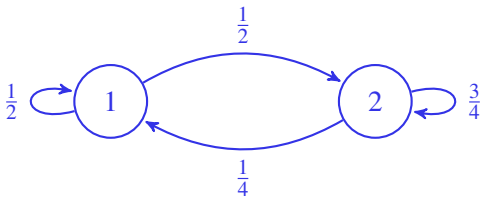
Consider a network of two webpages with connectivity matrix

$$\mathbf{C} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix},$$

where entry (i, j) is the fraction of page j's visitors who move to page i at each step. If $\mathbf{s}[n]$ holds the fraction of all visitors on each page at step n, then $\mathbf{s}[n+1] = \mathbf{C}\mathbf{s}[n]$.

(a) Draw the graph of this network. Use one node per webpage, and label each arrow with the fraction of visitors who follow it at each step.

Answer: Column j of $\mathbf{C}$ tells us where the visitors on page j go. From column 1, half of page 1's visitors stay on page 1 and half move to page 2. From column 2, a quarter of page 2's visitors move to page 1 and three quarters stay on page 2.



The weights on the arrows leaving each node sum to 1, which is the same as saying that each column of $\mathbf{C}$ sums to 1.

(b) Find the eigenvalues λ_1 and λ_2 of $\mathbf{C}$, and an eigenvector $\mathbf{v}_1$, $\mathbf{v}_2$ associated with each. What is the PageRank score vector $\mathbf{s}^*$?

Answer: The entries of $\mathbf{C}$ are nonnegative and each column sums to 1, so $\mathbf{C}$ is *column stochastic*. Every column stochastic matrix has an eigenvalue of 1, so $\lambda_1 = 1$.

For the second eigenvalue we use the trace. The trace of a matrix is the sum of its diagonal entries, and it also equals the sum of the eigenvalues:

$$\text{tr}(\mathbf{C}) = \tfrac{1}{2} + \tfrac{3}{4} = \tfrac{5}{4} = \lambda_1 + \lambda_2 = 1 + \lambda_2 \quad \Longrightarrow \quad \lambda_2 = \tfrac{1}{4}.$$

For $\lambda_1 = 1$:

$$(\mathbf{C} - \mathbf{I})\mathbf{v} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0} \quad \Longrightarrow \quad y = 2x, \qquad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

For $\lambda_2 = \frac{1}{4}$:

$$(\mathbf{C} - \tfrac{1}{4}\mathbf{I})\mathbf{v} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0} \quad \Longrightarrow \quad y = -x, \qquad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Any nonzero scalar multiples of these are also valid. The score vector is the eigenvector for $\lambda = 1$ scaled so that its entries sum to 1:

$$\mathbf{s}^* = \frac{1}{3}\mathbf{v}_1 = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}.$$

(c) Let $\mathbf{V} = \begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2\end{bmatrix}$ be the matrix whose columns are the eigenvectors, and let $\mathbf{\Lambda} = \begin{bmatrix}\lambda_1 & 0 \\ 0 & \lambda_2\end{bmatrix}$. Verify that $\mathbf{CV} = \mathbf{V\Lambda}$. Then compute $\mathbf{V}^{-1}$ and write $\mathbf{C}$ in its *diagonalized* form $\mathbf{C} = \mathbf{V\Lambda V}^{-1}$.

Answer: With $\mathbf{V} = \begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix}$ and $\mathbf{\Lambda} = \begin{bmatrix}1 & 0 \\ 0 & \frac{1}{4}\end{bmatrix}$,

$$\mathbf{CV} = \begin{bmatrix}\mathbf{Cv}_1 & \mathbf{Cv}_2\end{bmatrix} = \begin{bmatrix}\lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2\end{bmatrix} = \begin{bmatrix}1 & \frac{1}{4} \\ 2 & -\frac{1}{4}\end{bmatrix}, \qquad \mathbf{V\Lambda} = \begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix}\begin{bmatrix}1 & 0 \\ 0 & \frac{1}{4}\end{bmatrix} = \begin{bmatrix}1 & \frac{1}{4} \\ 2 & -\frac{1}{4}\end{bmatrix}.$$

The two agree. This is no accident: $\mathbf{CV} = \mathbf{V\Lambda}$ is just the two eigenvector equations $\mathbf{Cv}_i = \lambda_i\mathbf{v}_i$ stacked side by side, since multiplying $\mathbf{V}$ on the right by a diagonal matrix scales its columns.

Using the formula for the inverse of a 2×2 matrix, with $\det(\mathbf{V}) = (1)(-1) - (1)(2) = -3$,

$$\mathbf{V}^{-1} = \frac{1}{-3}\begin{bmatrix}-1 & -1 \\ -2 & 1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix}.$$

The inverse exists because $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent. Multiplying $\mathbf{CV} = \mathbf{V\Lambda}$ on the right by $\mathbf{V}^{-1}$,

$$\mathbf{C} = \mathbf{V\Lambda V}^{-1} = \begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix}\begin{bmatrix}1 & 0 \\ 0 & \frac{1}{4}\end{bmatrix}\cdot\frac{1}{3}\begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix}.$$

(d) After n steps the visitors are distributed as $\mathbf{s}[n] = \mathbf{C}^n\mathbf{s}[0]$. Use the diagonalized form to show that $\mathbf{C}^n = \mathbf{V\Lambda}^n\mathbf{V}^{-1}$, and use this to write each entry of $\mathbf{C}^n$ in terms of n.

Answer: Writing out the product, each $\mathbf{V}^{-1}$ meets a $\mathbf{V}$ and cancels:

$$\mathbf{C}^n = (\mathbf{V\Lambda V}^{-1})(\mathbf{V\Lambda V}^{-1})\cdots(\mathbf{V\Lambda V}^{-1}) = \mathbf{V\Lambda}(\mathbf{V}^{-1}\mathbf{V})\mathbf{\Lambda}(\mathbf{V}^{-1}\mathbf{V})\cdots\mathbf{\Lambda V}^{-1} = \mathbf{V\Lambda}^n\mathbf{V}^{-1}.$$

Powers of a diagonal matrix are easy: we just raise each diagonal entry to the n-th power, so $\mathbf{\Lambda}^n = \begin{bmatrix}1 & 0 \\ 0 & \left(\frac{1}{4}\right)^n\end{bmatrix}$. Then

$$\mathbf{C}^n = \begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix}\begin{bmatrix}1 & 0 \\ 0 & \left(\frac{1}{4}\right)^n\end{bmatrix}\cdot\frac{1}{3}\begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1 & \left(\frac{1}{4}\right)^n \\ 2 & -\left(\frac{1}{4}\right)^n\end{bmatrix}\begin{bmatrix}1 & 1 \\ 2 & -1\end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix}1+2\left(\frac{1}{4}\right)^n & 1-\left(\frac{1}{4}\right)^n \\ 2-2\left(\frac{1}{4}\right)^n & 2+\left(\frac{1}{4}\right)^n\end{bmatrix}.$$

As a check, $n = 1$ gives $\frac{1}{3}\begin{bmatrix}\frac{3}{2} & \frac{3}{4} \\ \frac{3}{2} & \frac{9}{4}\end{bmatrix} = \mathbf{C}$. Notice that we never had to multiply $\mathbf{C}$ by itself.

(e) What does $\mathbf{C}^n$ approach as $n \to \infty$? Use your answer to show that $\mathbf{s}[n]$ converges to the score vector $\mathbf{s}^*$ for *every* starting distribution $\mathbf{s}[0] = \begin{bmatrix}a \\ b\end{bmatrix}$, where $\mathbf{s}[0]$ is a nonnegative vector ($a, b \geq 0$) with $a + b = 1$.

Answer: In part (d) we found

$$\mathbf{C}^n = \frac{1}{3}\begin{bmatrix}1+2\left(\frac{1}{4}\right)^n & 1-\left(\frac{1}{4}\right)^n \\ 2-2\left(\frac{1}{4}\right)^n & 2+\left(\frac{1}{4}\right)^n\end{bmatrix}.$$

The only place n appears is in $\left(\frac{1}{4}\right)^n$, which goes to 0 as $n \to \infty$. Setting those terms to 0 in each entry leaves

$$\lim_{n\to\infty} \mathbf{C}^n = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{2}{3} \end{bmatrix}.$$

Each column of the limit is the score vector $\mathbf{s}^* = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$ from part (b). Multiplying this limit by the starting distribution,

$$\lim_{n\to\infty} \mathbf{s}[n] = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \frac{a+b}{3}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix} = \mathbf{s}^*,$$

no matter how the visitors were distributed at the start.

(f) Now consider a different network of two webpages in which each page links only to the other, so that its connectivity matrix is

$$\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The eigenvalues of $\mathbf{P}$ are $\lambda_1 = 1$ and $\lambda_2 = -1$. Diagonalize $\mathbf{P}$ and use the result to write each entry of $\mathbf{P}^n$ in terms of n. Does $\mathbf{P}^n$ approach a limit as $n \to \infty$? If every visitor starts on page 1, do the visitors settle into a steady distribution? Compare with what happened for $\mathbf{C}$.

Answer: For $\lambda_1 = 1$, $(\mathbf{P} - \mathbf{I})\mathbf{v} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. For $\lambda_2 = -1$, $(\mathbf{P} + \mathbf{I})\mathbf{v} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. So

$$\mathbf{V} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathbf{\Lambda} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \mathbf{V}^{-1} = \frac{1}{-2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and $\mathbf{P} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$. Just as in part (d), $\mathbf{P}^n = \mathbf{V}\mathbf{\Lambda}^n\mathbf{V}^{-1}$:

$$\mathbf{P}^n = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & (-1)^n \end{bmatrix} \cdot \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1+(-1)^n & 1-(-1)^n \\ 1-(-1)^n & 1+(-1)^n \end{bmatrix}.$$

This time the term that depends on n is $(-1)^n$, which does not go to 0: it flips between $+1$ and -1 forever. So $\mathbf{P}^n$ has no limit. It equals $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ when n is even and $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ when n is odd.

If every visitor starts on page 1, then $\mathbf{s}[n] = \mathbf{P}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is the first column of $\mathbf{P}^n$:

$$\mathbf{s}[n] = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ for even } n, \qquad \mathbf{s}[n] = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ for odd } n.$$

The visitors never settle: the whole crowd hops back and forth between the two pages. The network still has a score vector, $\mathbf{s}^* = \frac{1}{2}\mathbf{v}_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$, since $\mathbf{P}\mathbf{s}^* = \mathbf{s}^*$, but the visitors only sit at it if they start there.

The difference from $\mathbf{C}$ is the second eigenvalue. For $\mathbf{C}$, $\lambda_2 = \frac{1}{4}$ and $\left(\frac{1}{4}\right)^n \to 0$, so $\mathbf{C}^n$ converged and every starting distribution was pulled to the score vector. For $\mathbf{P}$, $|\lambda_2| = 1$, so that term never dies out. Diagonalizing the connectivity matrix shows us both things at once: the eigenvector for $\lambda = 1$ gives the score vector, and the remaining eigenvalues tell us whether the visitors actually converge to it.